# FIITJEE

# ALL INDIA TEST SERIES

## CONCEPT RECAPITULATION TEST – III

## JEE (Main)-2019

**Time Allotted: 3 Hours** **Maximum Marks: 360**

**General Instructions:**

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only one section: **Section-A**.

  **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

# Physics

## PART – I

### SECTION – A
### (One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A cyclist is moving with a speed of 6 $ms^{-1}$. As he approaches a circular turn on the road of radius 120 m, he applies brakes and reduces his speed at a constant rate of 0.4 $ms^{-2}$. The magnitude of the net acceleration of the cyclist on the circular turn just after brakes is
   (A) 0.5 $ms^{-2}$ (B) 1.0 $ms^{-2}$
   (C) 2.0 $ms^{-2}$ (D) 4.0 $ms^{-2}$

2. A force $\vec{F} = (3\hat{i} + 4\hat{j})$ Newton acts on a particle moving along a line 4y + kx = 3. The work done by the force is zero if the value of k is
   (A) 1 (B) 2
   (C) 3 (D) 4

3. Moment of inertia of uniform horizontal solid cylinder of mass M about an axis passing through its edge and perpendicular to the axis of the cylinder when its length is 6 times its radius R is
   (A) $\frac{39MR^2}{4}$ (B) $\frac{39MR}{4}$
   (C) $\frac{49MR}{4}$ (D) $\frac{49MR^2}{4}$

4. How many significant figures should be written in the sum 4.56g+9.032g+580.0078g+540.439g ?
   (A) 3 (B) 4
   (C) 5 (D) 6

5. A ball is thrown into the air and follows a parabolic trajectory. At the highest point in the trajectory
   (A) the velocity is zero, but the acceleration is not zero
   (B) both the velocity and the acceleration are zero
   (C) the acceleration is zero, but the velocity is not zero
   (D) neither the acceleration nor the velocity are zero

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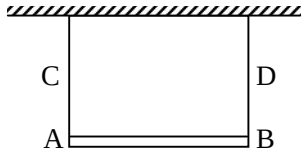
6. A thin string that can support a weight of 35.0 N, but breaks under any larger weight, is attached to the ceiling of an elevator. How large a mass can be attached to the string if the initial acceleration as the elevator starts to ascend is 3.20 m/s$^2$ ?
(A) 3.57 kg (B) 2.69 kg
(C) 2.96 kg (D) 5.30 kg

7. The potential energy (in joule) of a body of mass 2 kg moving in the x-y plane is given by U = 6x + 8y where the position coordinates x and y are measured in metre. If the body is at rest at point (6 m, 4m) at time t = 0, it will cross the y-axis at time t equal to
(A) 1 s (B) 2 s
(C) 3 s (D) 4 s

8. A shell is fired from a fixed cannon with a velocity 'v' at an angle $\theta$ with the horizontal direction. At the highest point on its path, it explodes into two pieces of equal mass. One of the pieces retraces its path to the cannon. The speed of the other piece immediately after the explosion is
(A) 3 v cos $\theta$ (B) 2 v cos $\theta$
(C) $\frac{3}{2}$ v cos $\theta$ (D) $\sqrt{\frac{3}{2}}$ v cos $\theta$

9. A body of mass *m* is projected with a velocity *u* at an angle of 60º with the horizontal. The magnitude of the angular momentum about the point of projection when the body is at the highest point of its trajectory is
(A) $\frac{mu^3}{2g}$ (B) $\frac{2mu^3}{5g}$
(C) $\frac{3mu^3}{16g}$ (D) $\frac{mu^3}{4g}$

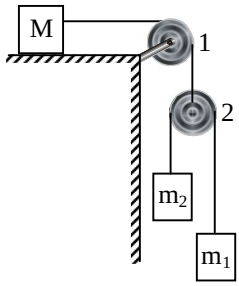
10. A uniform rod AB of mass m and length L is suspended by two strings C and D of negligible mass as shown in the figure. When string D is cut, the tension in string C will be




(A) $\frac{mg}{4}$ (B) mg
(C) 2 mg (D) 4 mg

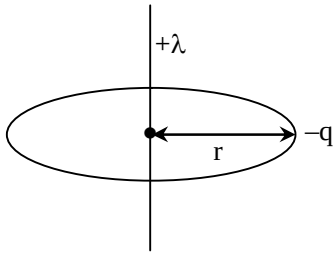
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11. In the arrangements shown in figure $m_1$ = 1kg, $m_2$ = 2 kg. Pulleys are massless and strings are light. For what value of M the mass $m_1$ moves with constant velocity (neglect friction)




(A) 6 kg (B) 4 kg
(C) 8 kg (D) 10 kg

12. A particle of charge –q and mass m moves in a circle of radius *r* around an infinitely long line charge of linear charge density +λ.




Then time period will be – (where k = $\frac{1}{4\pi \in_0}$)

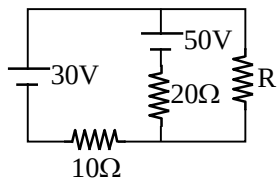
(A) $T = 2\pi r \sqrt{\frac{m}{2k\lambda q}}$

(B) $T^2 = \frac{4\pi^2 m}{2k\lambda q} r^3$

(C) $T = \frac{1}{2\pi r}\sqrt{\frac{2k\lambda q}{m}}$

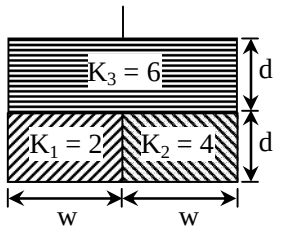
(D) $T = \frac{1}{2\pi r}\sqrt{\frac{m}{2k\lambda q}}$

13. In the circuit shown, the value of R in ohm that will result in no current through the 30 V battery, is




(A) 10Ω (B) 25Ω
(C) 30Ω (D) 40Ω

14. A parallel plate capacitor of capacitance C (without dielectrics) is filled by dielectric slabs as shown in figure. Then the new capacitance of the capacitor is



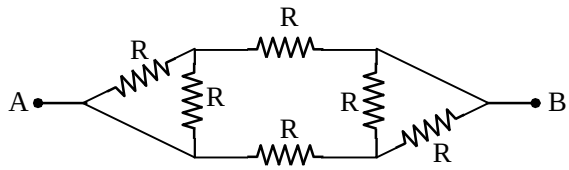

(A) 3.9 C (B) 4 C
(C) 2 C (D) 3 C

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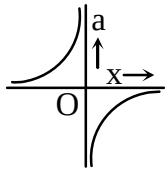
15. An LCR series circuit with 100Ω resistance is connected to an AC source of 200 V and angular frequency 300 radians per second. When only the capacitance is removed, the current lags behind the voltage by 60º. When only the inductance is removed, the current leads the voltage by 60º. Then the current and power dissipated in LCR circuit are respectively
(A) 1A, 200 watt (B) 1A, 400 watt
(C) 2A, 200 watt (D) 2A, 400 watt

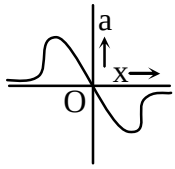
16. A wire of length λ is bent in the form of a circular coil of some turns. A current 'i' flow through the coil. The coil is placed in a uniform magnetic field B. The maximum torque on the coil can be
(A) $\frac{iB\ell^2}{4\pi}$ (B) $\frac{iB\ell^2}{\pi}$
(C) $\frac{iB\ell^2}{2\pi}$ (D) $\frac{2iB\ell^2}{\pi}$

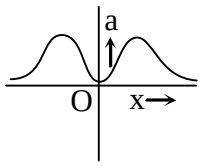
17. Equivalent resistance between A and B is
(A) $\frac{3}{4}R$
(B) $\frac{5}{3}R$
(C) $\frac{7}{5}R$
(D) R

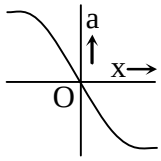



18. Two identical positive charges are fixed on the y-axis, at equal distances from the origin O. A particle with a negative charge starts on the negative x-axis at a large distance from O, moves along the x-axis, passed through O and moves far away from O. Its acceleration *a* is taken as positive along its direction of motion. The particle's acceleration *a* is plotted against its x-co-ordinate. Which of the following best represents the plot ?

(A) 


(B) 


(C) 


(D) 

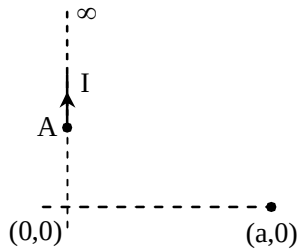

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19. A charged particle is projected with velocity $v_0$ along positive x-axis. The magnetic field B is directed along negative z-axis between x = 0 and x = L. The particle emerges out (at x = L) at an angle of 60º with the direction of projection. Find the velocity with which the same particle were projected (at x = 0) along positive x-axis so that when it emerges out (at x = L), the angle made by it is 30º with the direction of projection

(A) $2v_0$ (B) $\frac{v_0}{2}$

(C) $\frac{v_0}{\sqrt{3}}$ (D) $v_0\sqrt{3}$

20. An infinitely long wire carrying current I is along Y-axis such that its one end is at point A(0, b) while the wire extends upto $+\infty$. The magnitude of magnetic field strength at point (a, 0)




(A) $\frac{\mu_0 I}{4\pi a}\left(1+\frac{b}{\sqrt{a^2+b^2}}\right)$

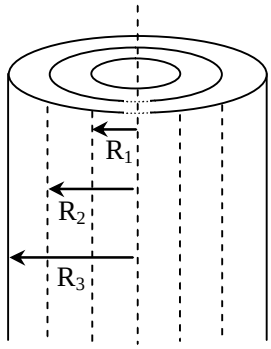
(B) $\frac{\mu_0 I}{4\pi a}\left(1-\frac{b}{\sqrt{a^2+b^2}}\right)$

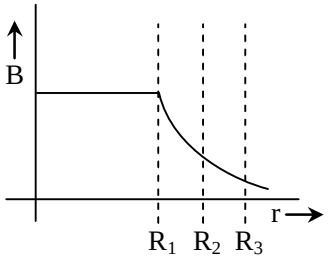
(C) $\frac{\mu_0 I}{4\pi a}\left(\frac{b}{\sqrt{a^2+b^2}}\right)$

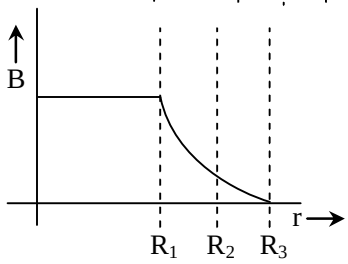
(D) None of these

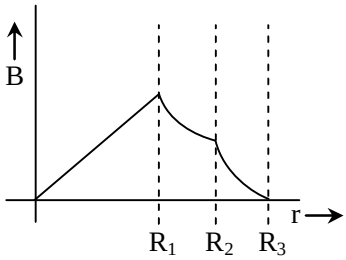
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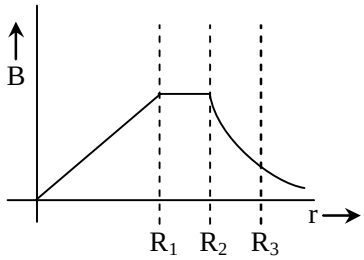
21. A coaxial cable is made up of two conductors. The inner conductor is solid and is of radius $R_1$ and the outer conductor is hollow of inner radius $R_2$ and outer radius $R_3$. The space between the conductors is filled with air. The inner and outer conductors are carrying currents of equal magnitudes and in opposite directions. Then the variation of magnetic field with distance from the axis is best plotted as



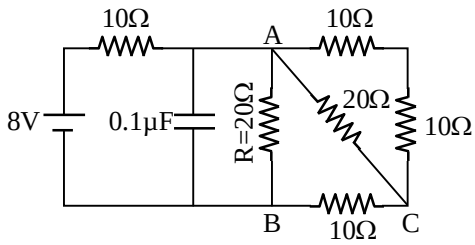

(A)



(B)



(C)



(D)



22. A capacitor of capacitance 0.1 μF is connected to a battery of emf 8V as shown in the figure. Under steady state condition
(A) Charge on the capacitor is 0.4 μC
(B) Charge on the capacitor is 0.2 μC
(C) Current in the resistor (R) between points A and B is 0.1 A
(D) Current in the resistor (R) between points A and B is 0.4 A



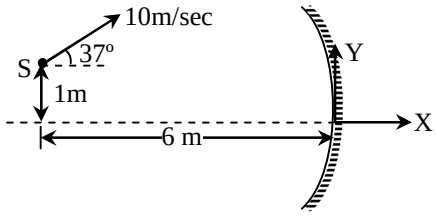

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23. In a sample of a radioactive substance what fraction of the initial number of nuclei will remain undecayed after time t=$\frac{T}{2}$, where T = half-life of radioactive substance

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2\sqrt{2}}$

(C) $\frac{1}{4}$ (D) $\frac{1}{\sqrt{2}-1}$

24. A point source S is moving with a speed of 10 m/sec in x-y plane as shown. The radius of curvature of concave mirror is 4m. What will be velocity vector of image formed by paraxial rays




(A) $-2\hat{i}-6\hat{j}$

(B) $-2\hat{i}-4\hat{j}$

(C) $-2\hat{i}-8\hat{j}$

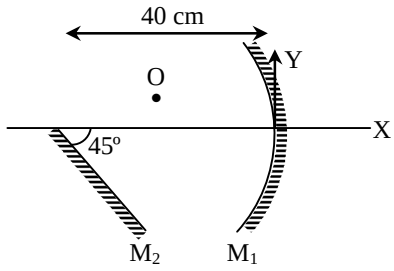
(D) $2\hat{i}-4\hat{j}$

25. Magnetic field at the centre (at Nucleus) of the hydrogen like atoms (atomic No. Z) due to motion of electron in $n^{th}$ orbit is proportional to

(A) $\frac{n^3}{Z^5}$ (B) $\frac{n^4}{Z}$

(C) $\frac{Z^2}{n^3}$ (D) $\frac{Z^3}{n^5}$

26. As shown in figure an object "O" is at the position (–10, 2) with respect to origin "P". The concave mirror $M_1$ has radius of curvature 30 cm. A place mirror $M_2$ is kept at a distance of 40 cm in front of concave mirror. Considering first reflection at concave mirror $M_1$ and second through plane mirror $M_2$. Find the co-ordinates of second image w.r.t. origin P

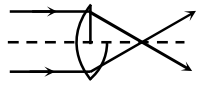


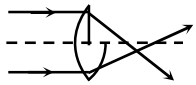

(A) –40, –70 (B) –52, –70

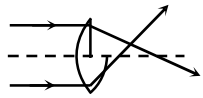
(C) – 46, –70 (D) –52, – 60

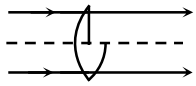
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27. Choose the correct ray diagram of a thin equi-convex lens which is cut as shown in the figure.

(A) 

(B) 

(C) 

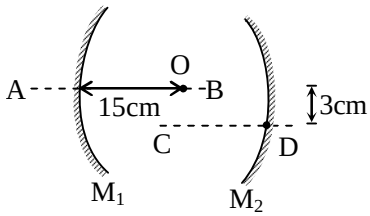
(D) 

28. A radioactive isotope is being produced at a constant rate X. Half life of the radioactive substance is Y. After some time the number of radioactive nuclei constant. The value of the constant is

(A) $\frac{XY}{\ell n(2)}$ (B) XY

(C) XY $\ell$ n(2) (D) $\frac{X}{Y}$

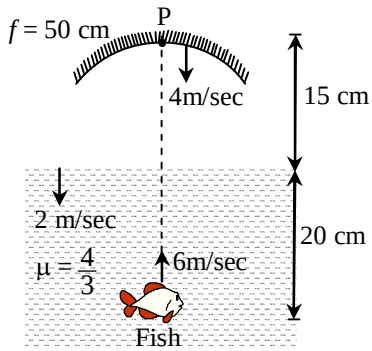
29. In the shown figure $M_1$ and $M_2$ are two concave mirrors of same focal length 10 cm. AB and CD are their principal axis respectively. A point object O is kept on the line AB at a distance 15 cm from $M_1$. The distance between the mirrors is 20 cm. Considering two successive reflections first on $M_1$ and then on $M_2$. The distance of final image from the line AB is




(A) 3 cm (B) 1.5 cm

(C) 4.5 cm (D) 1 cm

30. In the figure shown a concave mirror of focal length 50 cm is made to move downward with constant velocity of 4 m/sec. The level of water having $\mu = \frac{4}{3}$ in also falling downward at speed of 2 m/sec. Inside the water the fish is moving upward with speed of 6 m/sec and is lying along principal axis of concave mirror. Find out the speed of image of fish which is formed after reflection through concave mirror at the instant shown




(A) 36 m/sec downward

(B) 54 m/sec downward

(C) 18 m/sec downward

(D) none of these

***Space for Rough work***

# Chemistry

## PART – II

### SECTION – A
### (One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. A definite amount of solid $NH_4HS$ is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. $NH_4HS$ decomposes to give $NH_3$ and $H_2S$ and at equilibrium total pressure in flask is 0.84 atm. The equilibrium constant for the reaction is :
(A) 0.30 (B) 0.18
(C) 0.17 (D) 0.11

32. $Mg_2C_3 + H_2O \longrightarrow$ X (organic compound). Compound X is
(A) $C_2H_2$ (B) $CH_4$
(C) propyne (D) ethene

33. $BeCl_2 + LiAlH_4 \longrightarrow X + LiCl + AlCl_3$
(A) X is LiH (B) X is $BeH_2$
(C) X is $BeCl_2 \cdot 2H_2O$ (D) None

34. 1 cc of 0.1 N HCl is added to 99 cc solution of NaCl. The pH of the resulting solution will be
(A) 7 (B) 3
(C) 4 (D) 1

35. 10 ml of $\frac{M}{200} H_2SO_4$ is mixed with 40 ml of $\frac{M}{200} H_2SO_4$. The pH of the resulting solution is
(A) 1 (B) 2
(C) 2.3 (D) none of these

36. The incorrect statement(s) is/are
(A) Mg cannot form complexes
(B) Be can form complexes due to a very small atomic size
(C) the first ionisation potential of Be is higher than that of Mg
(D) Mg forms an alkaline hydroxide while Be forms amphoteric oxides

***Space for Rough work***

37. $MgBr_2$ and $MgI_2$ are soluble in acetone because of
(A) Their ionic nature
(B) Their coordinate nature
(C) Their metallic nature
(D) Their covalent nature

38. aq. NaOH + $P_4$ (white) $\longrightarrow$ $PH_3$ + X; compound X is
(A) $NaH_2PO_2$
(B) $NaHPO_4$
(C) $Na_2CO_3$
(D) $NaHCO_3$

39. $H_3BO_3 \xrightarrow{T_1} X \xrightarrow{T_2} Y \xrightarrow{red\ hot} B_2O_3$ if $T_1 < T_2$ then X and Y respectively are
(A) X = Metaboric acid and Y = Tetraboric acid
(B) X = Tetraboric acid and Y = Metaboric acid
(C) X = Borax and Y = Metaboric acid
(D) X = Tetraboric acid and Y = Borax

40. $RCl \xrightarrow[Si]{cu-powder} R_2SiCl_2 \xrightarrow{H_2O} R_2Si(OH)_2 \xrightarrow{condensation} A$
Compound (A) is
(A) a linear silicone
(B) a chlorosilane
(C) a linear silane
(D) a network silane

41. The initial rate of zero order reaction of the gaseous reaction A(g) $\longrightarrow$ 2B(g) is $10^{-2}$ M $min^{-1}$. If the initial concentration of A is 0.1 M, what would be the concentration of B after 60 sec.?
(A) 0.09 M
(B) 0.01 M
(C) 0.02 M
(D) 0.002 M

42. The vapour density of $N_2O_4$ at a certain temperature is 30. What is the % dissociation of $N_2O_4$ at this temperature?
(A) 53.3%
(B) 106.6%
(C) 26.7%
(D) None

43. For the reaction : 2HI(g) $\rightleftharpoons$ $H_2(g)$ + $I_2(g)$, the degree of dissociated (a) of HI(g) is related to equilibrium constant $K_p$ by the expression
(A) $\frac{1+2\sqrt{K_p}}{2}$
(B) $\sqrt{\frac{1+2K_p}{2}}$
(C) $\sqrt{\frac{2K_p}{1+2K_p}}$
(D) $\frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$

***Space for Rough work***

44. Solution of sodium metal in liquid ammonia is a strong reducing agent due to presence of
(A) solvated sodium ions (B) solvated hydrogen ions
(C) sodium atoms or sodium hydroxide (D) solvated electrons

45. The solubility of anhydrous $AlCl_3$ and hydrous $AlCl_3$ in diethyl ether are $S_1$ and $S_2$ respectively. Then
(A) $S_1 = S_2$ (B) $S_1 > S_2$
(C) $S_1 < S_2$ (D) $S_1 < S_1$ but not $S_1 = S_2$

46. For the reaction A + B $\longrightarrow$ C; starting with different initial concentration of A and B, initial rate of reaction were determined graphically in four experiments.

| S.No. | $[A]_0$/M (Initial conc.) | $[B]_0$/M (Initial conc.) | rate/(M $sec^{-1}$) |
|---|---|---|---|
| 1 | $1.6 \times 10^{-3}$ | $5 \times 10^{-2}$ | $10^{-3}$ |
| 2 | $3.2 \times 10^{-3}$ | $5 \times 10^{-2}$ | $4 \times 10^{-3}$ |
| 3 | $1.6 \times 10^{-3}$ | $10^{-1}$ | $2 \times 10^{-3}$ |
| 4 | $3.2 \times 10^{-3}$ | $10^{-1}$ | $8 \times 10^{-3}$ |

Rate law for reaction from above data is
(A) $r = k[A]^2 [B]^2$ (B) $r = k[A]^2 [B]$
(C) $r = k[A] [B]^2$ (D) $r = k[A] [B]$

47. The compound which contains ionic as well as covalent bonds is
(A) $C_2H_4Cl_2$ (B) $CH_3I$
(C) KCN (D) $H_2O_2$

48. The hybridisation and shape of $BrF_3$ molecule are :
(A) $sp^3d$ and bent *T* shape (B) $sp^2d^2$ and tetragonal
(C) $sp^3d$ and bent (D) none of these

49. The correct order of decreasing bond angle is :
(A) $H_2O > Cl_2O > F_2O$ (B) $Cl_2O > H_2O > F_2O$
(C) $F_2O > Cl_2O > H_2O$ (D) $F_2O > H_2O > Cl_2O$

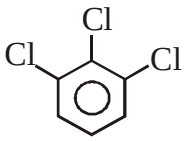
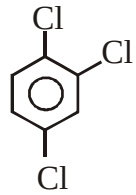
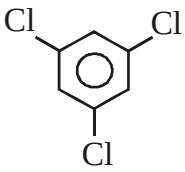
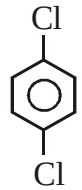
50. The electron affinity of the members of oxygen family of the periodic table, follows the sequence
(A) O > S > Se (B) S > O < Se
(C) O < S > Se (D) Se > O > S

***Space for Rough work***

51. The correct order of atomic / ionic size
(A) N < Li < B
(B) Cl < Mg < Ca
(C) $Ca^{+2} < S^{-2} < Cl^{-}$
(D) $Na^{+} < Mg^{+2} < Cl^{-}$

52. If radius of second stationary orbit (in Bohr's atom) is R. Then radius of third orbit will be
(A) R/3
(B) 9R
(C) R/9
(D) 2.25R

53. The shortest wavelength of $He^{+}$ ion in Balmer series is x, then longest wavelength in the Paschen series of $Li^{+2}$ is
(A) $\frac{36x}{5}$
(B) $\frac{16x}{7}$
(C) $\frac{9x}{5}$
(D) $\frac{5x}{9}$

54. Which has maximum dipole moment.
(A) 
(B) 
(C) 
(D) 

55. The correct order in which the O—O bond length increases in the following :
(A) $H_2O_2 < O_2 < O_3$
(B) $O_3 < H_2O_2 < O_2$
(C) $O_2 < H_2O_2 < O_3$
(D) $O_2 < O_3 < H_2O_2$

56. Which among the following orbital diagram violates Pauli's exclusion principle, Hund's rule as well as Aufbau's rule :
(A) 
(B) 
(C) 
(D) 

***Space for Rough work***

57. The increasing order of acidic nature of $Li_2O$, BeO, $B_2O_3$

(A) $Li_2O > BeO < B_2O_3$ (B) $Li_2O < BeO < B_2O_3$

(C) $Li_2O < BeO > B_2O_3$ (D) $Li_2O > BeO > B_2O_3$

58. Order of hydration energy

(A) $Na^+ = Mg^{2+} = Al^{3+}$ (B) $Na^+ > Mg^{2+} > Al^{3+}$

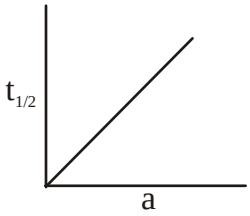
(C) $Al^{+3} > Mg^{2+} > Na^+$ (D) $Al^{+3} > Mg^{2+} < Na^+$

59. For the reaction : $CO(g) + \frac{1}{2}O_2(g) \text{ l } CO_2(g)$, $K_p / K_c$ is :

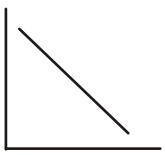
(A) RT (B) $(RT)^{-1}$

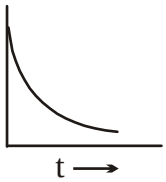
(C) $(RT)^{-1/2}$ (D) $(RT)^{1/2}$

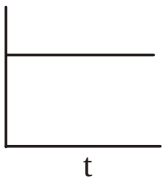
60. Consider the reaction A $\longrightarrow$ B, graph between half life ($t_{1/2}$) and initial concentration (a) of the reactant is

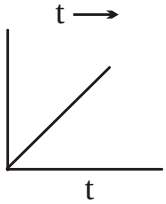



Hence graph between $-\frac{d[A]}{dt}$ and time will be

(A) $-\frac{d[A]}{dt}$ 

(B) $-\frac{d[A]}{dt}$ 


(C) $-\frac{d[A]}{dt}$ 


(D) $-\frac{d[A]}{dt}$ 


***Space for Rough work***

# Mathematics

## PART – III

### SECTION – A
### (One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let $f(x) = (x-1)^{100}.(x-2)^{2(99)}.(x-3)^{3(98)}..... (x-100)^{100}$. The value of $\frac{f'(101)}{f(101)}$ will be

(A) 5051 (B) 5050
(C) 5049 (D) 0

62. Let F(x) be a polynomial function of degree 2018 and leading coefficient unity such that F(0) = 2017, F(a) = 2016, F(b) = 2015, ....F(2017) = 0. The value of F(2018) is of the form n! – a, where n, a $\in$ N and a is least possible value, then n + a is equal to

(A) 2016 (B) 2017
(C) 2019 (D) 2021

63. If $\omega$ is a complex cube root of unity and $f(n) = 2(1+\omega)(1+\omega^2) + 3(2+\omega)(2+\omega^2)+....(n+1)(n+\omega)(n+\omega^2)$, then the value of $f(19)$ is

(A) 36150 (B) 36100
(C) 36101 (D) 36119

64. Let z $\in$ C satisfy the equation $|z^2| + 2(z+\bar{z}) - 5 = 0$, then the complex number z + 3 + 2i will lie on

(A) circle with center 1 – 2i and radius 3 (B) circle with center 3 – 2i and radius 4
(C) circle with center 1 + 2i and radius 3 (D) circle with center 3 + 2i and radius 4

65. Let $f(x)$ be a derivable function defined over the set of real numbers such that $f(f(x)) = \lambda(x^7 + 2x)$, $\lambda \neq 0$, then $\forall$ x $\in$ R, $f(x)$ is always

(A) Increasing (B) Decreasing
(C) Non monotonic (D) Either increasing or decreasing

66. Consider 12 face diagonals of a cubical block. How many pairs of them are skew lines?

(A) 30 (B) 15
(C) 45 (D) 60

***Space for Rough work***

67. Logical equivalent of $p \rightarrow (p \vee q)$ is

(A) $\sim p \wedge \sim q$ (B) $\sim p \rightarrow (p \wedge q)$

(C) $q \rightarrow (p \rightarrow q)$ (D) $p \rightarrow \sim q$

68. For 200 observations of a specific set of data, the mean was found to be 48 and the standard deviation is 3. Sum of squares of these 200 observations will be

(A) 462000 (B) 462600

(C) 482600 (D) 462800

69. Number of values of x for which $\cos^{-1}x + \cot^{-1}2x = -\pi/2$ is

(A) 4 (B) 2

(C) 1 (D) 0

70. If $A = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$; $AA^T = I$, xyz = 1 and tr.(A) > 0, then the value of $x^3 + y^3 + z^3$ is

(where $A^T$, tr.(A) denotes transpose of square matrix A and sum of elements of principal diagonal of matrix A, and I is identity matrix of order 3)

(A) 3xyz (B) x + y + z

(C) 0 (D) 4

71. Least positive integral value of x for which $x^2 - 4x > \cot^{-1}x$ is

(A) 1 (B) 2

(C) 5 (D) 6

72. If x + y = k is a tangent to the hyperbola $x^2 - 2y^2 = 18$, then sum of squares of possible values of k is

(A) 3 (B) 9

(C) 12 (D) 18

73. Consider $f(x) = \sin x \ \forall \ x \in \left[0, \frac{\pi}{2}\right]$; $f(x) + f(\pi - x) = 2 \ \forall \ x \in \left(\frac{\pi}{2}, \pi\right]$ and $f(x) = f(2\pi - x) \ \forall \ x \in (\pi, 2\pi)$. If n, m denotes number of points where $f(x)$ is discontinuous and non derivable respectively in $[0,2\pi)$, then value of n + m is

(A) 0 (B) 1

(C) 2 (D) 4

***Space for Rough work***

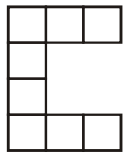
74. Let $f : R \to R$ and $g : R \to R$ be two functions such that $f(x) + f''(x) = -x.g(x).f'(x)$ and $g(x) > 0$, $f'(x) > 0 \ \forall \ x \in R$, then for $h(x) = f^2(x) + (f'(x))^2$
(A) x = 0 is point of maxima (B) x = 0 is point of minima
(C) h(x) is monotonic increasing at x = 0 (D) h(x) is monotonic decreasing at x = 0

75. If $\alpha + \sin\beta = 2018$; $\alpha + 2018\cos\beta = 2017$, $\beta \in \left(0, \frac{\pi}{2}\right]$, then value of $[\alpha + \beta]$ is
(where [.] denotes greatest integer function)
(A) 2016 (B) 2017
(C) 2018 (D) 2020

76. Common tangents of $9x^2 - 9y^2 = 8$ and $y^2 = 32x$ passes through (a,0), then a is
(A) $\frac{8}{9}$ (B) $-\frac{8}{9}$
(C) $\frac{9}{8}$ (D) $-\frac{9}{8}$

77. For different values of $\lambda$, the circle $x^2 + y^2 + (8 + \lambda)x + (8 + \lambda)y + 16 + 12\lambda = 0$ passes through two fixed points A and B. Value of $\lambda$ for which tangents at A and B to the circle intersect at origin, is
(A) –8 (B) $\frac{8}{3}$
(C) –16 (D) $\frac{16}{3}$

78. Six different letters are arranged in boxes shown in figure. Each box can hold exactly one letter. Number of ways in which letters can be arranged so that none of the rows remain empty, is



(A) 7200 (B) 9600
(C) 10000 (D) 10800

79. A and B alternately throw a pair of die until one of them wins. A is considered winner if he gets a total 7 and B is considered winner, if he gets a total of 5. If A starts the game, then probability that one of the die shows face 3 on the last throw is
(A) $\frac{11}{28}$ (B) $\frac{9}{28}$
(C) $\frac{5}{28}$ (D) $\frac{3}{28}$

***Space for Rough work***

80. If $n \in N$ and $(1 - 2x + 5x^2 + 10x^3)(1 + x)^n = a_0 + a_1x + a_2x^2 + .....$, and $a_1^2 = 2a_2$, then value of $a_0 + n$ is
(A) 1 (B) 5
(C) 6 (D) 7

81. Let be three non zero, non coplanar vectors and be three vectors given by $\vec{\alpha} = \vec{a} - 4\vec{b} - 2\vec{c}$, $\vec{\beta} = \vec{a} + \vec{b} - 2\vec{c}$, $\vec{\gamma} = 3\vec{a} + 2\vec{b} + \vec{c}$. If $V_1$ is volume of parallelopiped with coterminous edges and $V_2$ is volume of tetrahedron with coterminous edges $\vec{a}, \vec{b}, \vec{c}$, then $\frac{V_1}{V_2}$ is
(A) 1 (B) 6
(C) 35 (D) 210

82. Angles of elevation of top of a tower when observed from ground floor and roof of a building of height h are $\alpha$ and $\beta$ respectively, then height of tower will be
(A) $\frac{h\cos\beta}{\cot\beta - \cot\alpha}$ (B) $\frac{h\cot\beta}{\cot\beta - \cot\alpha}$
(C) $\frac{h\cot\alpha}{\cot\beta - \cot\alpha}$ (D) $\frac{h\cos\alpha}{\cot\beta - \cot\alpha}$

83. If the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{1} = \frac{y-6}{2} = \frac{z+\lambda}{2}$ intersect, then value of $\lambda$ is
(A) –9 (B) 9
(C) 8 (D) –8

84. Solution of differential equation given by $\frac{dy}{dx} = \frac{x^2 + y + 1}{x(1 - x\sin y)}$, where y(1) = 0, is
(A) $y + x\cos y = x^2 + x - 1$ (B) $y + x\cos y = x^2 - x + 1$
(C) $y + x\cos y = -x^2 + x + 1$ (D) $y + x\cos y = x^2 + x + 1$

85. If roots of equation $(k^2 - 4k + 5)x^2 + (2k - 1)x - 3k = 0$ lie on either sides of unity then number of integral values of k is
(A) 1 (B) 2
(C) 3 (D) 4

***Space for Rough work***

86. Value of $\lim_{x\to 1} x^{\frac{1}{\ell nx}}$, is

(A) $\frac{1}{e}$ (B) $e^2$

(C) e (D) does not exist

87. If $\cot(\theta + \alpha)$, $3\cot\theta$, $\cot(\theta - \alpha)$ are in AP, then value of $\frac{\sin^2\theta}{\sin^2\alpha}$, wherever defined, is

(A) $\frac{3}{2}$ (B) 3

(C) $\frac{2}{3}$ (D) 2

88. For an ellipse, the locus of mid points of chords which are drawn through an end of minor axis will be

(A) a parabola (B) an ellipse

(C) a hyperbola (D) a pair of lines

89. Value of $I = \int_{-\pi/2}^{\pi/2} \frac{e^x dx}{\left(1+e^x\right)\left(1+\tan^2 x\right)}$ is

(A) $\frac{\pi}{2e}$ (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{4e}$ (D) $\frac{\pi}{4}$

90. Area enclosed between the curves given by $y = x^4$ and $y = x^2 + 12$ will be (in sq. units)

(A) $\frac{304}{15}$ (B) $\frac{594}{15}$

(C) $\frac{608}{15}$ (D) cannot be determined

***Space for Rough work***